UNIT4

Working of Naïve Bayes' Classifier:

Problem: If the weather is sunny, then the Player should play or not?

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Sl No	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Frequency table for the Weather Conditions:

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	4

Bayesian Classification

Sl No	Outlook	Play
0	Rainy	Yes
1	Sunny	Yes
2	Overcast	Yes
3	Overcast	Yes
4	Sunny	No
5	Rainy	Yes
6	Sunny	Yes
7	Overcast	Yes
8	Rainy	No
9	Sunny	No
10	Sunny	Yes
11	Rainy	No
12	Overcast	Yes
13	Overcast	Yes

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	4

Weather	No	Yes	
Overcast	0	5	5/14= 0.35
Rainy	2	2	4/14=0.29
Sunny	2	3	5/14=0.35
All	4/14=0.29	10/14=0.71	

Applying Bayes'theorem: P(A|

P(Yes|Sunny) = P(Sunny | Yes) * P(Yes) / P(Sunny)
P(Sunny | Yes)= 3/10= 0.3
P(Sunny)= 0.35
P(Yes)=0.71
So,
P(Yes|Sunny) = 0.3*0.71/0.35= 0.60

P(No|Sunny) = P(Sunny|No) *P(No)/P(Sunny)
P(Sunny|NO) = 2/4=0.5
P(No)= 0.29
P(Sunny)= 0.35
So
P(No|Sunny) = 0.5*0.29/0.35 = 0.41

So
as we can see from the above calculation that
P(Yes|Sunny)>P(No|Sunny)
Hence on a Sunny day, Player can play the game.

K-NEAREST NEIGHBOUR (KNN)

- Unlike other algorithms, KNN doesn't "learn" a model during training; it simply stores the data and waits until you give it a new point to classify.

- Similar things usually exist in close proximity.
- **Analogy:** If you want to know if a person is a "Sports Fan," look at their 3 closest friends. If 2 out of 3 are sports fans, that person likely is too.

K-NEAREST NEIGHBOUR (KNN

- KNN classifies data based on nearest neighbors.
- Steps:
- Choose K
- Calculate distance
- Take majority class
- Distance used:
  Euclidean distance

PROBLEM

- Classifying a "Mystery Beverage" as **Soda** or **Fruit Juice** based on **Sugar Content** and **Acid Level**.

Beverage	Sugar (g)	Acid (1-10)	Label
Drink A	12	7	Soda
Drink B	10	8	Soda
Drink C	2	3	Fruit Juice
Drink D	3	2	Fruit Juice

- **New Data Point (X):** Sugar = **6g**, Acid = **4**.
- **Task:** Classify X using **K=3.**

Step 1: Calculate the Distance

Teach students to use the **Euclidean Distance** formula. It is simply the Pythagorean theorem in a coordinate plane.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- **Distance to A:** $\sqrt{(12-6)^2 + (7-4)^2} = \sqrt{36+9} = \sqrt{45} \approx \mathbf{6.7}$
- **Distance to B:** $\sqrt{(10-6)^2 + (8-4)^2} = \sqrt{16+16} = \sqrt{32} \approx \mathbf{5.6}$
- **Distance to C:** $\sqrt{(2-6)^2 + (3-4)^2} = \sqrt{16+1} = \sqrt{17} \approx \mathbf{4.1}$
- **Distance to D:** $\sqrt{(3-6)^2 + (2-4)^2} = \sqrt{9+4} = \sqrt{13} \approx \mathbf{3.6}$

Step 2: Ranking the Neighbors

Sort the distances from smallest to largest to find who is "closest."

1. **3.6** (Drink D - Fruit Juice)
2. **4.1** (Drink C - Fruit Juice)
3. **5.6** (Drink B - Soda)
4. 6.7 (Drink A - Soda)

Step 3: Majority Voting

Since $K = 3$, we select the top 3 neighbors:

- **2 votes** for Fruit Juice (C and D)
- **1 vote** for Soda (B)

Final Result: The Mystery Beverage is classified as **Fruit Juice**.

BASICS OF PREDICTION

- Prediction is a fundamental task in data mining that involves using historical **data** to forecast unknown or future values.
- Regression Analysis
- Regression analysis is a statistical method used to examine the relationship between a dependent variable and one or more independent variables. It is widely applied in research to predict outcomes, identify trends, and understand the strength and nature of relationships between variables.

BASICS OF PREDICTION

- Types of Regression
- **Simple Linear Regression**: Examines the relationship between one dependent and one independent variable. It is commonly used in weather forecasting and financial analysis.
- **Multiple Linear Regression**: Involves two or more independent variables to predict a dependent variable. It is used in fields like real estate and public healthcare

MODEL EVALUATION METRICS

- Confusion Matrix
- confusion matrix helps assess classification model performance in machine learning by comparing predicted values against actual values for a dataset.
- A Confusion matrix is an N x N matrix used for evaluating the performance of a classification model, where N is the total number of target classes.

MODEL EVALUATION METRICS

- For a binary classification problem, we would have a **2 x 2 matrix,** as shown below, with 4 values:

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

- The target variable has two values: **Positive** or **Negative**
- The **columns** represent the **actual values** of the target variable
- The **rows** represent the **predicted values** of the target variable

True Positive (TP)

- The predicted value matches the actual value, or the predicted class matches the actual class.
- The actual value was positive, and the model predicted a positive value.

True Negative (TN)

- The predicted value matches the actual value, or the predicted class matches the actual class.
- The actual value was negative, and the model predicted a negative value.

False Positive (FP)

- The predicted value was falsely predicted.
- The actual value was negative, but the model predicted a positive value.

False Negative (FN)

- The predicted value was falsely predicted.
- The actual value was positive, but the model predicted a negative value.

- True Positive (TP) = 560, meaning the model correctly classified 560 positive class data points.
- True Negative (TN) = 330, meaning the model correctly classified 330 negative class data points.
- False Positive (FP) = 60, meaning the model incorrectly classified 60 negative class data points as belonging to the positive class.
- False Negative (FN) = 50, meaning the model incorrectly classified 50 positive class data points as belonging to the negative class.

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
PREDICTED VALUES	POSITIVE	560	60
	NEGATIVE	50	330

- **Accuracy**
- **Definition:**
- Accuracy measures the **overall proportion of correct predictions** (both TP and TN) out of all predictions.
- $Accuracy = \frac{TP+TN}{Total}$

- **Precision**
- **Definition:**
- Precision (also called "positive predictive value") measures the **reliability of positive predictions**: of all cases predicted as positive, how many are actually positive.
- $Precision = \frac{TP}{TP+FP}$

- ***Recall*** is also known as ***sensitivity*** or ***true positive rate*** and is defined as follows:
- $Recall = \frac{TP}{TP+FN}$

- **F1 Score**
- $F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$
- Used when:

  Precision & Recall both important.

- **Given Confusion Matrix**
- Suppose:
- TP = 40

  TN = 50

  FP = 10

  FN = 20
- Total = 120

- **Calculate Accuracy**
- $Accuracy = \frac{40+50}{120}$
- $Accuracy = \frac{90}{120} = 0.75$
- Accuracy = 75%

- **Calculate Precision**
- $Precision = \frac{40}{40+10}$
- $Precision = \frac{40}{50} = 0.8$
- Precision = 80%
- **Calculate Recall**
- $Recall = \frac{40}{40+20}$
- $Recall = \frac{40}{60} = 0.67$
- Recall = 67%**Interpretation**
- Model correctly predicts 75% cases.
- When model predicts positive → 80% correct.
- It detects 67% of actual positive cases.